



INDIAN INSTITUTE
OF TECHNOLOGY
PALAKKAD

Electronic Door Lock System

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Team

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Submitted To

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1. Overview

The proposed project aims to design and develop an Electronic Door Lock System using an Arduino Uno, where users can input a password through a keypad, view the entered data on an LCD display, and control access to a door using an electronic lock.

2. Objective

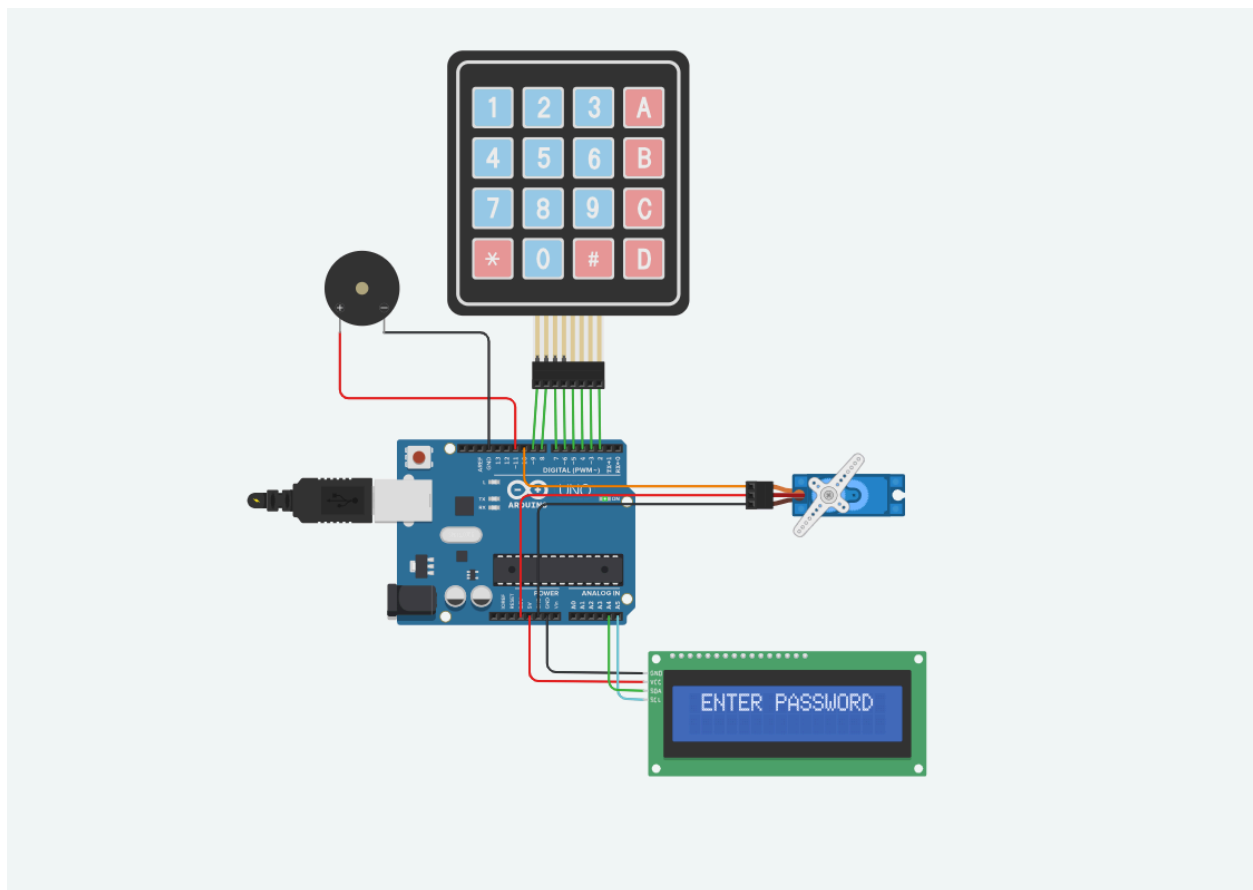
1. A functional electronic door lock system that unlocks with a correct password.
2. Real-time password entry displayed on an LCD screen.
3. Audio (buzzer) feedback for access status.
4. A demonstrable prototype that enhances understanding of Arduino-based security systems.

3. Components Used

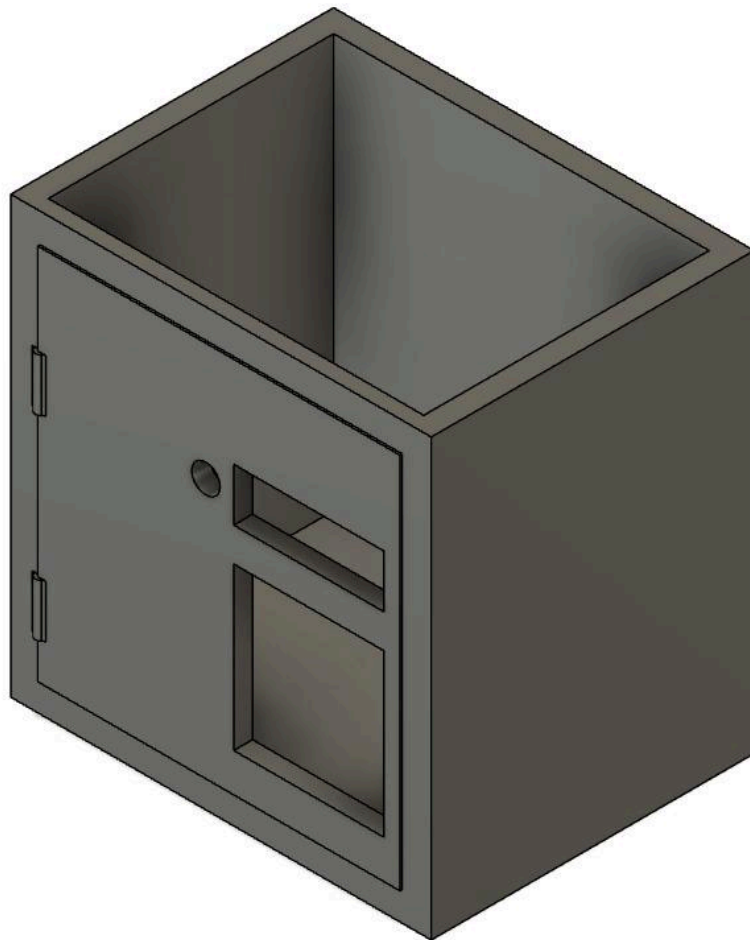
1. Arduino UNO
2. LCD & I2C Module
3. Servo Motor
4. 5V Buzzer
5. 4x4 Keypad
6. Jumper Wires
7. Door Latch
8. Wood MDF Board Sheets

4. Circuit Design

The following is the design of the Project



5. 3D Design



*Not to scale

6. Process of the Operation

Arduino Uno R3: The microcontroller will act as the brain of the system to process inputs and control the lock.

4x4 Keypad: Used for entering the password.

16x2 LCD Display: Displays the entered password and system messages (e.g., "Access Granted," "Access Denied").

Servo Motor: Acts as the locking/unlocking mechanism for the door.

Buzzer: Provides audio feedback for incorrect entries or successful access.

7. Practical Usage

1. Home and office security systems.
2. Locker systems.
3. Access control in restricted areas.
4. Educational projects for learning about embedded systems and Arduino programming.

8. Budget

Components	Quantity	Estimated Cost(INR)
Arduino Uno	1	1200
4x4 Keypad	1	400
16x2 LCD Display	1	800
Servo Motor	1	800
5V Buzzer	1	200
Jumper Wires	-	150
Casing Material (MDF)	-	800
Miscellaneous	-	300

Total		4650*
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*Total Budget may vary

9. References

1. Simple circuit Door Lock by **Simple Circuits**

▶ How to make password door lock arduino project | Arduino Project

[Simple circuit Door Lock by Simple Circuits](#)

▶ Password Door Lock Using Arduino | Arduino Project | Tinkercad Tutorial

[TinkerCad Tutorial by Simple Circuits](#)

2. Arduino Code by **Simple Circuits**

[Arduino Code by Simple Circuits](#)